

Section 4: Physical chemistry

a) Acids, alkalis and salts

b) Energetics

c) Rates of reaction

d) Equilibria

a) Acids, alkalis and salts

Students will be assessed on their ability to:

- 4.1 describe the use of the indicators litmus, phenolphthalein and methyl orange to distinguish between acidic and alkaline solutions
- 4.2 understand how the pH scale, from 0–14, can be used to classify solutions as strongly acidic, weakly acidic, neutral, weakly alkaline or strongly alkaline
- 4.3 describe the use of universal indicator to measure the approximate pH value of a solution
- 4.4 define acids as sources of hydrogen ions, H^+ , and alkalis as sources of hydroxide ions, OH^-
- 4.5 predict the products of reactions between dilute hydrochloric, nitric and sulfuric acids; and metals, metal oxides and metal carbonates (excluding the reactions between nitric acid and metals)
- 4.6 recall the general rules for predicting the solubility of salts in water:
 - i all common sodium, potassium and ammonium salts are soluble
 - ii all nitrates are soluble
 - iii common chlorides are soluble, except silver chloride
 - iv common sulfates are soluble, except those of barium and calcium
 - v common carbonates are insoluble, except those of sodium, potassium and ammonium
- 4.7 describe how to prepare soluble salts from acids
- 4.8 describe how to prepare insoluble salts using precipitation reactions
- 4.9 describe how to carry out acid-alkali titrations

b) Energetics

Students will be assessed on their ability to:

- 4.10 recall that chemical reactions in which heat energy is given out are described as exothermic and those in which heat energy is taken in are endothermic
- 4.11 describe simple calorimetry experiments for reactions, such as combustion, displacement, dissolving and neutralisation in which heat energy changes can be calculated from measured temperature changes
- 4.12 calculate molar enthalpy change from heat energy change**
- 4.13 understand the use of ΔH to represent molar enthalpy change for exothermic and endothermic reactions
- 4.14 represent exothermic and endothermic reactions on a simple energy level diagram
- 4.15 recall that the breaking of bonds is endothermic and that the making of bonds is exothermic
- 4.16 use average bond energies to calculate the enthalpy change during a simple chemical reaction**

c) Rates of reaction

Students will be assessed on their ability to:

- 4.17 describe experiments to investigate the effects of changes in surface area of a solid, concentration of solutions, temperature and the use of a catalyst on the rate of a reaction
- 4.18 describe the effects of changes in surface area of a solid, concentration of solutions, pressure of gases, temperature and the use of a catalyst on the rate of a reaction
- 4.19 understand the term 'activation energy' and represent it on a reaction profile
- 4.20 explain the effects of changes in surface area of a solid, concentration of solutions, pressure of gases and temperature on the rate of a reaction in terms of particle collision theory
- 4.21 understand that a catalyst speeds up a reaction by providing an alternative pathway with lower activation energy

d) Equilibria

Students will be assessed on their ability to:

- 4.22 recall that some reactions are reversible and are indicated by the symbol $\rightleftharpoons$ in equations
- 4.23 describe reversible reactions such as the dehydration of hydrated copper(II) sulfate and the effect of heat on ammonium chloride
- 4.24 understand the concept of dynamic equilibrium
- 4.25 predict the effects of changing the pressure and temperature on the equilibrium position in reversible reactions